

Foreground (How to Describe an Abnormality)

Waveforms in the foreground may be artifact or cerebral activity (normal variants or abnormalities).

Each type of waveform observed may be identified and classified based on three key features:

1. *Location*: Is the waveform predominantly diffuse (generalized), hemispheric (lateralized), focal (regional), or multifocal?
2. *Occurrence*: Is the waveform sporadic (single or abundant but occurs randomly) or repetitive (recurs in rhythmic or periodic pattern)?
3. *Morphology*: Is it pointed (spike or sharp) or blunt (slow wave) in appearance?

For example, the term generalized periodic discharge (GPD) denotes a sharp or spike wave (discharge) that is generalized in location and occurs repetitively in a periodic fashion as shown in Figure 11.1.

These key terms are simply introduced here, they are explained in the following chapters.

After you have described a waveform based on the above three key features of location, occurrence, and morphology, use the following additional modifiers to qualify your description [1]:

1. prevalence
2. duration
3. amplitude
4. frequency

The appropriate usage of each of these terms is explained below. They are particularly useful in describing abnormalities seen on continuous EEG monitoring in the critically ill patients [1]. It is important to note that these modifiers are used in context of the entire record or epoch and not an individual page of the EEG.

Prevalence

This denotes the approximate amount (percentage) of the specific waveform in the record:



Figure 11.1 Sixty-two-year-old man in a coma showing occipital-predominant generalized periodic discharges (GPDs).

1. continuous, if the waveform occurs contiguously through almost the entire record.
2. intermittent, if the waveform occurs in parts during the record. Intermittent waveforms may be further described as abundant (>50%), frequent (49%–10%), occasional (9%–1%), or rare (<1%), depending on approximately how often they are observed during the entire record [1].

Figure 11.1 shows “continuous” generalized periodic discharges, while Figure 11.2 shows “intermittent” discharges.

Duration

This is the typical length of time for which - intermittent patterns observed in the record, this is particularly useful in continuous EEG monitoring. Intermittent patterns may be further described as very long (hours), long (minutes), or brief (seconds), depending on their approximate duration [1].

For example, Figure 11.2 shows a very brief (1 s) burst of frontally predominant generalized spike wave discharges.

Amplitude

Measure the amplitude (absolute amplitude) from the peak to the trough of the waveform (positive to negative peak) and not peak to baseline. It is best to measure the amplitude in longitudinal bipolar montage (in the channel it is best seen). Figure 11.3 shows the correct method to measure the amplitude of a waveform.

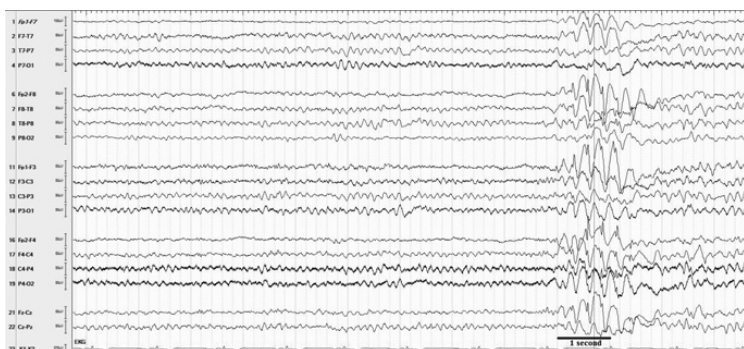


Figure 11.2 Thirty-four-year-old man with epilepsy showing intermittent very brief (1 s) frontally predominant generalized spike wave discharges (GSW).

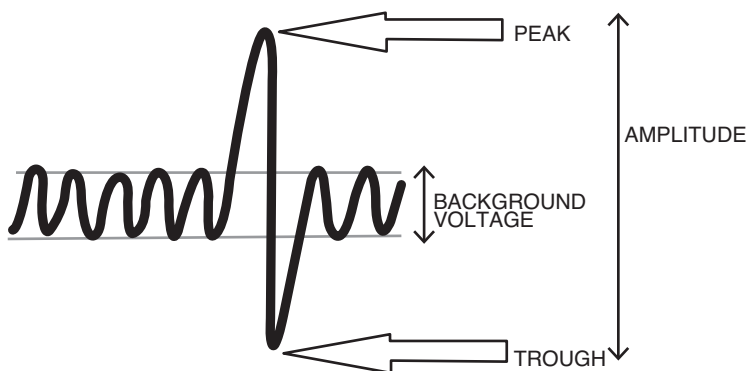


Figure 11.3 Correct method to measure amplitude of a waveform.

The typical amplitude can then be described as low (less than 20 μV), medium (less than 50 μV), or high (more than 200 μV) [1].

Frequency

The rate (typical and range) should be specified for a pattern to the nearest 0.5/s division [1]. For example, in Figure 11.2, the discharges typically occur at a frequency of 4 Hz (four times per second).

Reporting the Foreground Abnormality

When reporting the waveform, the modifiers (prevalence, duration, amplitude, and frequency) should precede the three key features of location, occurrence, and morphology.

For example, Figure 11.1 shows “continuous, long duration, high-amplitude, 2–1 Hz, occipitally predominant generalized periodic discharges” and Figure 11.2 shows “intermittent, very brief, high-amplitude, 4 Hz, frontally predominant generalized spike wave discharges.”

Chapter Summary

1. First identify the foreground abnormality (or activity) based on three key features of location (generalized, lateralized, focal, or multifocal), occurrence (sporadic or repetitive), and morphology (sharp or blunt).
2. Next, qualify your description with the modifiers of prevalence, duration, amplitude, and frequency.
3. When reporting the waveform, the modifiers (prevalence, duration, amplitude, and frequency) should precede the three key features of location, occurrence, and morphology.

Reference

1. Hirsch LJ, LaRoche SM, Gaspard N, et al. American clinical neurophysiology society’s standardized critical care EEG terminology: 2012 version. *Journal of Clinical Neurophysiology*. 2013 Feb 1;30(1):1–27.